

NBE Projects Proposal

Idea 1: Family Banking

Problem Statement

Parents often struggle to monitor their children's spending, regulate their monthly allowance, and enforce restrictions on transaction amounts and spending categories.

Key Features

- Monthly allowance management.
- Parent approval for transactions exceeding a configurable spending threshold.
- Specialized and customizable spending bundles.
- Multi-child support.
- Savings goal tracker.

Idea 2: Smart Bank Routing

Problem Statement

Customers may travel to an ATM only to find that it is out of service or out of cash, resulting in wasted time and frustration.

Key Features

- Route customers to the nearest fully operational ATM or branch.
- Dashboard displaying ATM status, supported transaction types, and cash availability.

Idea 3: Senior Remote Protection

Problem Statement

Senior citizens are particularly vulnerable to online financial scams and fraudulent transactions, often leading to financial losses and reduced trust in digital banking services.

Key Features

- A secondary authentication request is sent to a designated guardian before high-risk transactions are completed.

Idea 4: Intelligent Bill Splitting Platform

Problem Statement

Splitting shared expenses is often more complicated than simply dividing the total amount equally among participants. Different situations require different splitting rules; for example, hotel costs depend on the number of nights each guest stayed, restaurant bills depend on the meals each person ordered, and recurring household expenses such as rent, utilities, and subscriptions follow fixed or weighted distributions. Manually tracking these expenses frequently leads to confusion, calculation errors, and difficulty determining who owes whom. As expenses accumulate over time, identifying the minimum number of transactions required to settle all outstanding debts becomes increasingly challenging.

Key Features

- Multiple expense splitting methods (equal, per-night, per-item, weighted, percentage-based).
- Recurring expense management for rent, utilities, subscriptions, and other shared bills.
- Debt simplification algorithm to minimize the number of transactions required to settle balances.
- OCR-based receipt scanning with AI-assisted parsing to automatically extract purchased items and prices.
- Interactive item assignment interface allowing each participant to claim or remove receipt items before finalizing the split.